RANK

The Rank–Nullity Theorem

Pivot columns and free columns share the n columns of A.

THEOREM

For an $m \times n$ matrix A: $\operatorname{rank}(A) + \operatorname{nullity}(A) = n$.

1 **COUNT THE COLUMNS**

A 3×5 matrix: $n = 5$ unknowns in $A\mathbf{x} = \mathbf{0}$.

2 **RANK**

Rank 2: two pivot columns, a column space that is a plane in $\mathbb{R}^3$.

3 **NULLITY**

$5 - 2 = 3$: three free columns, a three-dimensional null space in $\mathbb{R}^5$.

4 **SQUARE CASE**

For $n \times n$: rank n gives nullity 0 and A invertible; rank below n gives a nontrivial null space, determinant 0, A singular.

SUBTRACTING THE RANK FROM THE ROWS

Using the number of rows gives a 2×4 matrix of rank 2 a nullity of $2 - 2 = 0$. But $A\mathbf{x} = \mathbf{0}$ has 4 unknowns, so the nullity is $4 - 2 = 2$.